

RF and Microwave Engineering Numerical Companion

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · Chapter 6, every amplifier S-parameter set and the 82 Bh filter, 2069–2082 BS

6. RF Design Practices: Numerical Problems

299 numerical marks, 14.4% of the archive, 61% of this chapter · 10 solved problems covering every amplifier S-parameter set in 22 papers and the 82 Bh filter · 13 transistor sets, 1 filter design

READ THIS FIRST

Every amplifier question in the archive is one procedure run on four numbers. The papers attach the formula sheet (Δ , K , μ , B_1 , C_1 , Γ_S , Γ_L , C_S , R_S , C_L , R_L , $G_{T,max}$), so the marks are in doing the steps in order and getting the complex arithmetic right.

- **The procedure, always in this order**, always tabulated: (1) magnitudes and products, (2) Δ , (3) K and μ , verdict, (4) if stable, B_1 , C_1 , B_2 , C_2 , Γ_S , Γ_L , $G_{T,max}$, (5) unilateral G_S , G_0 , G_L , $G_{TU,max}$, (6) figure of merit M and the error bound, (7) compare. Problem 1 shows every line.
- If $K < 1$ **stop and say so**. The simultaneous conjugate match does not exist (the square root goes imaginary or $|\Gamma| > 1$). Draw stability circles instead and give the unilateral gain with its design points checked to be stable (Problems 3 to 5).
- **Work in polar form for products, rectangular for sums**. Most slips are a sum done in polar form. Keep four significant figures until the end.
- **“Compare” means a sentence**: which is larger, by how many dB, and whether M says the unilateral figure can be trusted.
- Every number below is from `src/amp.py`, which cross-checks $G_{T,max}$ against $|S_{21}/S_{12}|(K - \sqrt{K^2 - 1})$, confirms $\Gamma_S = \Gamma_{in}^*$ and $\Gamma_L = \Gamma_{out}^*$, and walks each matching network forward.

Problem 1: Stability and maximum gain, bilateral and unilateral (the full method) 6+3+3 8+4 15

71 Ma, 81 Ba, 82 Ba · the same transistor in three papers; 71 Bh prints it with S_{11} at $+150^\circ$ (Problem 2)

QUESTION AS PRINTED 71 Ma [15]

Using the following S-parameters of $S_{11}=0.55\angle 150^\circ$, $S_{12}=0.04\angle 20^\circ$, $S_{21}=2.82\angle 180^\circ$ and $S_{22}=0.45\angle 30^\circ$, calculate and compare maximum power gain for unilateral and bilateral modes.

81 Ba [6+3+3] asks the same plus a design flowchart (§6.4); 82 Ba [8+4] asks to “design a microwave amplifier for maximum power gain” for both cases (add the matching networks of Step 6).

Step 1: magnitudes and products

Quantity	Working	Value
$ S_{11} ^2, S_{22} ^2$	$0.55^2, 0.45^2$	0.3025, 0.2025
$ S_{21} ^2$	2.82^2	7.9524
$S_{11}S_{22}$	$0.55 \times 0.45\angle(-150^\circ - 30^\circ)$	$0.2475\angle 180^\circ = -0.2475$
$S_{12}S_{21}$	$0.04 \times 2.82\angle(20^\circ + 180^\circ)$	$0.1128\angle -160^\circ = -0.1060 - j0.0386$

Step 2 and 3: Δ , K , μ

- $\Delta = S_{11}S_{22} - S_{12}S_{21} = -0.2475 - (-0.1060 - j0.0386) = -0.1415 + j0.0386 = \boxed{0.1467\angle 164.7^\circ}$, $|\Delta|^2 = 0.0215$.
- $K = \frac{1 - 0.3025 - 0.2025 + 0.0215}{2 \times 0.1128} = \frac{0.5165}{0.2256} = \boxed{2.290}$.
- $\Delta S_{11}^* = 0.1467\angle 164.7^\circ \times 0.55\angle 150^\circ = 0.0807\angle -45.3^\circ$; $S_{22} - \Delta S_{11}^* = 0.3728\angle -26.7^\circ$; $\mu = \frac{1 - 0.3025}{0.3728 + 0.1128} = \boxed{1.436}$.
- $K > 1$, $|\Delta| < 1$ (and $\mu > 1$) $\Rightarrow$ **unconditionally stable**: any passive source and load are safe, so the conjugate match may be used.

Step 4: bilateral maximum gain (simultaneous conjugate match)

Quantity	Formula and substitution	Value
B_1	$1 + S_{11} ^2 - S_{22} ^2 - \Delta ^2 =$ $1 + 0.3025 - 0.2025 - 0.0215$	1.0785
C_1	$S_{11} - \Delta S_{22}^*$	$0.4866 \angle -148.0^\circ$
B_2	$1 + S_{22} ^2 - S_{11} ^2 - \Delta ^2$	0.8785
C_2	$S_{22} - \Delta S_{11}^*$	$0.3728 \angle -26.7^\circ$
Discriminant	$B_1^2 - 4 C_1 ^2 = 1.1632 - 0.9471$ (same for B_2, C_2)	0.2159, root 0.4646
Γ_S	$\frac{B_1 - \sqrt{\cdot}}{2C_1} = \frac{1.0785 - 0.4646}{2 \times 0.4866 \angle -148.0^\circ}$	$0.6307 \angle 148.0^\circ$
Γ_L	$\frac{B_2 - \sqrt{\cdot}}{2C_2} = \frac{0.8785 - 0.4646}{2 \times 0.3728 \angle -26.7^\circ}$	$0.5551 \angle 26.7^\circ$
$G_{T,max}$	$\frac{1}{1 - \Gamma_S ^2} S_{21} ^2 \frac{1 - \Gamma_L ^2}{ 1 - S_{22}\Gamma_L ^2} =$ $1.6606 \times 7.9524 \times 1.2275$	$16.21 = 12.10$ dB
Check	$\frac{ S_{21} }{ S_{12} } (K - \sqrt{K^2 - 1}) =$ $70.5 \times (2.290 - 2.060)$	16.21 ✓

Take the minus sign because $B_1, B_2 > 0$; the plus sign gives $|\Gamma| > 1$.

Step 5 and 6: unilateral maximum gain and its validity

Quantity	Formula and substitution	Value
$G_{S,max}$	$1/(1 - S_{11} ^2) = 1/0.6975$	1.434 = 1.56 dB
G_0	$ S_{21} ^2$	7.952 = 9.00 dB
$G_{L,max}$	$1/(1 - S_{22} ^2) = 1/0.7975$	1.254 = 0.98 dB
$G_{TU,max}$	$1.434 \times 7.952 \times 1.254$	$14.30 = 11.55$ dB
M	$\frac{ S_{11} S_{12} S_{21} S_{22} }{(1 - S_{11} ^2)(1 - S_{22} ^2)}$ $= \frac{0.55 \times 0.04 \times 2.82 \times 0.45}{0.6975 \times 0.7975}$	0.0502
Error bound	$\frac{1}{(1 + M)^2} < \frac{G_T}{G_{TU}} < \frac{1}{(1 - M)^2}$ $\frac{1}{(1.0502)^2} < \frac{G_T}{G_{TU}} < \frac{1}{(0.9498)^2}$	-0.43 to +0.45 dB

Step 7: compare

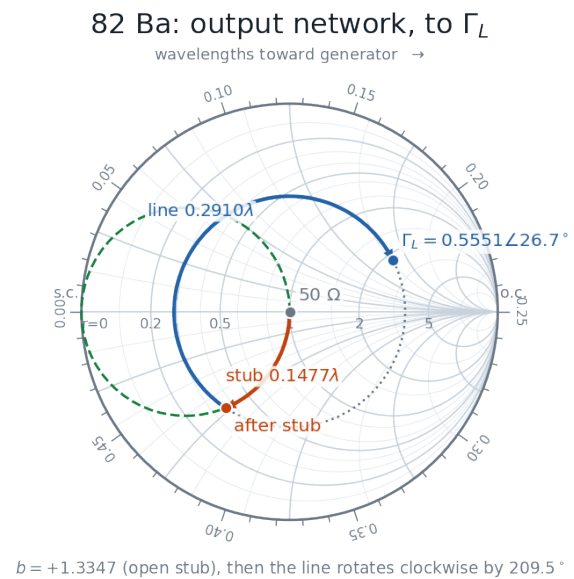
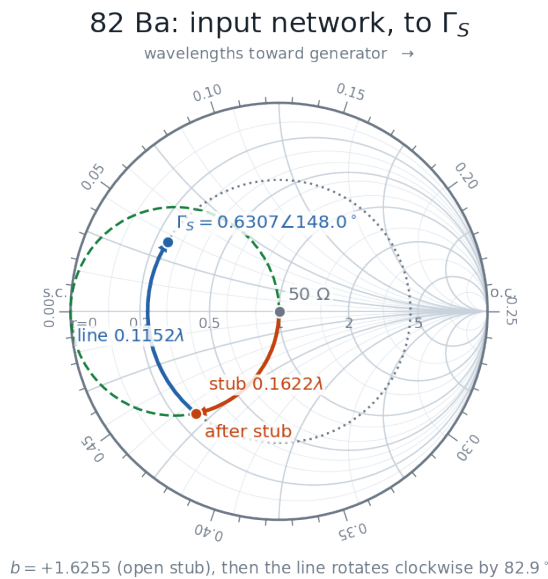
- $G_{T,max} = 12.10$ dB (bilateral) vs $G_{TU,max} = 11.55$ dB (unilateral): the bilateral design gives **0.55 dB more**.
- The unilateral error bound is ± 0.45 dB, and the true bilateral figure lies just outside it: the unilateral assumption is **approximately valid** ($M = 0.05$ is at the usual limit) but slightly pessimistic, because S_{12} feedback here adds gain.

Design (82 Ba): matching networks for the bilateral case

- **Job of each network:** turn a $50\ \Omega$ termination into the Γ the transistor needs. From the gate looking back to the source it must see $\Gamma_S = 0.6307 \angle 147.96^\circ$; from the drain looking to the load, $\Gamma_L = 0.5551 \angle 26.74^\circ$ (Step 4).
- **Stub first:** at the $50\ \Omega$ end $y = 1$, the chart centre. A shunt stub adds only susceptance, $y = 1 + jb$, so the point slides along the $g = 1$ circle. Choose b so it lands on the circle $|\Gamma| = |\Gamma_S|$.
- **Line second:** a lossless $50\ \Omega$ line keeps $|\Gamma|$ and only rotates it clockwise (away from the stub, towards the device) by $2\beta l$, i.e. 720° per λ . It fixes the angle.
- **Finding b :** $\Gamma = \frac{1 - y}{1 + y} = \frac{-jb}{2 + jb} \Rightarrow |\Gamma|^2 = \frac{b^2}{4 + b^2} \Rightarrow b = \frac{2|\Gamma|}{\sqrt{1 - |\Gamma|^2}}$, with $\angle \Gamma = -90^\circ - \tan^{-1}(b/2)$.
- **Lengths:** open stub $b = \tan \beta l_s \Rightarrow l_s = \frac{\tan^{-1} b}{360^\circ} \lambda$; line $l = \frac{\angle \Gamma_{stub} - \angle \Gamma_{target}}{720^\circ} \lambda$ (add 360° to the numerator if negative).

Step	Input network (Γ_S)	Output network (Γ_L)
$ \Gamma , \Gamma ^2$	0.6307, 0.3978	0.5551, 0.3081
$\sqrt{1 - \Gamma ^2}$	0.7760	0.8318
b	$2 \times 0.6307/0.7760 = 1.6255$	$2 \times 0.5551/0.8318 = 1.3347$
Open stub	$\tan^{-1} 1.6255 = 58.40^\circ$; $58.40/360 = \mathbf{0.1622\lambda}$	$\tan^{-1} 1.3347 = 53.16^\circ$; 0.1477λ
Γ after stub	$-90^\circ - \tan^{-1} 0.8128: 0.6307 \angle -129.10^\circ$	$-90^\circ - \tan^{-1} 0.6673: 0.5551 \angle -123.72^\circ$
Rotation needed	$-129.10 - (147.96 - 360) = 82.94^\circ$	$-123.72 - (26.74 - 360) = 209.54^\circ$
Line	$82.94^\circ/720 = \mathbf{0.1152\lambda}$	$209.54^\circ/720 = \mathbf{0.2910\lambda}$

- **Circuit:** $50\ \Omega$ source $\rightarrow$ shunt open stub $0.1622\lambda \rightarrow 0.1152\lambda$ line $\rightarrow$ transistor $\rightarrow 0.2910\lambda$ line $\rightarrow$ shunt open stub $0.1477\lambda \rightarrow 50\ \Omega$ load. Stubs sit at the $50\ \Omega$ ends; lines touch the transistor.
- $-b$ also works (input stub 0.3378λ , line 0.4738λ ; output 0.3523λ , 0.1347λ); $+b$ gives the shorter network on both sides. Same 12.10 dB either way.



The table on the chart: the stub slides the $50\ \Omega$ centre along $g = 1$ (green) onto the $|\Gamma|$ circle (dotted); the line rotates it clockwise to Γ_S or Γ_L . Drawn by `ampfig.py`, which walks each network forward to confirm it reaches Γ and stops if any tabled step differs.

Problem 2: The same method on the other unconditionally stable sets 10 12 18 4+4+4+4 5+5 7+8

69 Bh, 70 Bh, 71 Bh, 72 Ma, 74 Bh, 74 Ma, 77 Ch, 80 Bh, 80 Ch, 81 Bh, 82 Bh · identical steps; answers tabulated with the intermediates a marker checks

QUESTION AS PRINTED 80 Bh [10]

A GaAs MESFET has the following S-parameters measured with a $50\ \Omega$ resistance at 10 GHz: $S_{11} = 0.55 \angle -160^\circ$, $S_{12} = 0.04 \angle 10^\circ$, $S_{21} = 4.82 \angle 180^\circ$, $S_{22} = 0.45 \angle -30^\circ$. Determine the stability and compare maximum power gains for unilateral and bilateral modes using supplied formulas.

Other statements, verbatim in the Sorted PYQ document: 70 Bh “find the maximum gain” [10]; 74 Bh adds a flowchart and “trace Z_{in} , Z_{out} on the Smith chart” [4+4+4+4]; 69 Bh “design an amplifier to attain maximum gain at 4.0 GHz” [18]; 77 Ch “Ku-band satellite transmit amplifier, HEMT at 4 GHz” [5+5]; 80 Ch “check the stability and find the unilateral gain” [7+8]; 72 Ma “check the stability and find the maximum power gain” [10]; 74 Ma [12] and 82 Bh [12] “stability and compare”; 81 Bh “investigate with neat flow diagram the stability” [3+8]; 71 Bh as Problem 1 with $S_{11} = 0.55 \angle 150^\circ$.

Stability

Paper	$S_{11}, S_{12}, S_{21}, S_{22}$	Δ	K	μ	Verdict
80 Bh	0.55 $\angle$ -160, 0.04 $\angle$ 10, 4.82 $\angle$ 180, 0.45 $\angle$ -30	0.094 $\angle$ 125.2	1.306	1.147	uncond. stable
70 Bh, 74 Bh	0.656 $\angle$ 146.7, 0.122 $\angle$ 46.1, 2.30 $\angle$ 44.7, 0.172 $\angle$ -117.1	0.247 $\angle$ -65.6	1.071	1.080	uncond. stable (barely)
69 Bh, 77 Ch	0.72 $\angle$ -116, 0.03 $\angle$ 57, 2.60 $\angle$ 76, 0.73 $\angle$ -54	0.488 $\angle$ -162.3	1.195	1.040	uncond. stable
82 Bh	0.65 $\angle$ 146, 0.12 $\angle$ 46, 2.30 $\angle$ 44, 0.17 $\angle$ -172	0.339 $\angle$ -73.0	1.202	1.317	uncond. stable
80 Ch	0.45 $\angle$ -160, 0.04 $\angle$ 20, 3.12 $\angle$ 180, 0.35 $\angle$ -40	0.101 $\angle$ 107.7	2.746	1.776	uncond. stable
72 Ma (79 Ch)	0.45 $\angle$ 163, 0.04 $\angle$ 40, 2.55 $\angle$ -106, 0.46 $\angle$ -65	0.306 $\angle$ 103.3	3.332	1.877	uncond. stable
74 Ma	0.78 $\angle$ -113, 0.028 $\angle$ 247, 2.60 $\angle$ 76, 0.81 $\angle$ -54	0.681 $\angle$ -171.7	1.367	1.097	uncond. stable
81 Bh	0.6 $\angle$ -140, 0.03 $\angle$ 62, 2.40 $\angle$ 54, 0.70 $\angle$ -58	0.374 $\angle$ 170.0	2.011	1.161	uncond. stable
71 Bh (as printed)	0.55 $\angle$ +150, 0.04 $\angle$ 20, 2.82 $\angle$ 180, 0.45 $\angle$ -30	0.254 $\angle$ 94.0	2.479	1.574	uncond. stable

Bilateral and unilateral maximum gain

Paper	Γ_S	Γ_L	$G_{T,max}$	G_S, G_0, G_L (dB)	M	$G_{TU,max}, \text{bound}$
80 Bh	0.736 $\angle$ 156.7	0.683 $\angle$ 25.0	17.49 dB	1.56, 13.66, 0.98	0.086	16.21 dB , -0.71/+0.78
70 Bh, 74 Bh	0.850 $\angle$ -150.4	0.655 $\angle$ 76.3	11.13 dB	2.44, 7.23, 0.13	0.057	9.81 dB , -0.48/+0.51
69 Bh, 77 Ch	0.872 $\angle$ 123.4	0.876 $\angle$ 61.0	16.71 dB	3.17, 8.30, 3.31	0.182	14.78 dB , -1.45/+1.75
82 Bh	0.743 $\angle$ -150.0	0.378 $\angle$ 88.9	10.11 dB	2.38, 7.23, 0.13	0.054	9.75 dB , -0.46/+0.49
80 Ch	0.503 $\angle$ 156.2	0.419 $\angle$ 33.6	11.68 dB	0.98, 9.88, 0.57	0.028	11.43 dB , -0.24/+0.25
72 Ma (79 Ch)	0.401 $\angle$ -160.6	0.413 $\angle$ 67.2	9.91 dB	0.98, 8.13, 1.03	0.034	10.15 dB , -0.29/+0.30
74 Ma	0.752 $\angle$ 101.9	0.789 $\angle$ 45.2	16.06 dB	4.07, 8.30, 4.64	0.342	17.01 dB , -2.55/+3.63
81 Bh	0.699 $\angle$ 146.1	0.772 $\angle$ 61.7	13.28 dB	1.94, 7.60, 2.92	0.093	12.47 dB , -0.77/+0.84
71 Bh (as printed)	0.582 $\angle$ -156.4	0.490 $\angle$ 19.3	11.72 dB	1.56, 9.00, 0.98	0.050	11.55 dB , -0.43/+0.45

- **Reading the comparison:** bilateral exceeds unilateral in every set except 72 Ma / 79 Ch and 74 Ma, where the S_{12} feedback **subtracts** gain.
- **74 Ma and 69 Bh / 77 Ch:** $M = 0.34$ and 0.18 , bounds of several dB: the unilateral model is **not** acceptable for these transistors; quote the bilateral figure as the design gain.
- **70 Bh / 74 Bh:** $K = 1.07$, only just stable, and $|\Gamma_S| = 0.85$: a high-Q, narrow-band input match.
- **71 Bh:** the other three papers print S_{11} at -150° ; the $+150^\circ$ is probably a misprint. Both give the same unilateral gain (magnitudes only) but differ by 0.38 dB bilaterally; say which one you used.
- **74 Bh, trace Z_{in} and Z_{out} :** at the conjugate match $\Gamma_{in} = \Gamma_S^* = 0.850\angle 150.4^\circ \Rightarrow Z_{in} = 4.32 + j13.13 \Omega$ ($z = 0.086 + j0.263$, near the short-circuit end, upper half); $\Gamma_{out} = \Gamma_L^* = 0.655\angle -76.3^\circ \Rightarrow Z_{out} = 25.5 - j56.9 \Omega$ ($z = 0.51 - j1.14$, lower half, capacitive).
- **81 Bh** asks stability only: stop after the first table, and draw the flowchart of §6.4.

Problem 3: A conditionally stable transistor 5+5 10

70 Ma, 72 Ash, 80 Ba, 79 Bh · $S_{11} = 0.894\angle -60.6^\circ$: the most repeated single set

QUESTION AS PRINTED 72 Ash [5+5]

Justify that a transistor having following S-parameters $S_{11} = 0.894\angle -60.6^\circ$, $S_{12} = 0.020\angle 62.4^\circ$, $S_{21} = 3.122\angle 123.6^\circ$ and $S_{22} = 0.781\angle -27.6^\circ$ is conditionally stable while designing an amplifier. Considering unilateral model calculate maximum gain.

70 Ma [10] asks only the justification; 80 Ba [5+5] “investigate the stability” at 6 GHz; 79 Bh [5+5] “determine the stability and compare maximum power gains for bilateral and unilateral modes”.

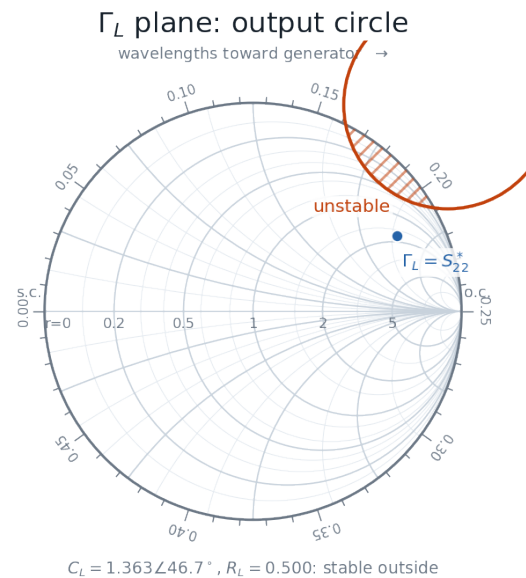
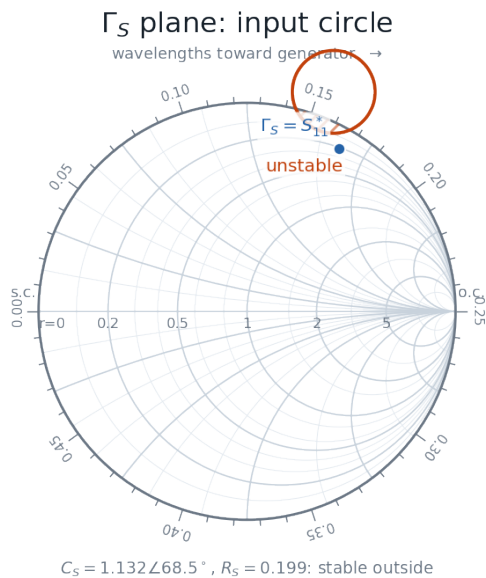
Step 1 to 3: stability

- $|S_{11}|^2 = 0.7992$, $|S_{22}|^2 = 0.6100$, $|S_{21}|^2 = 9.747$, $|S_{12}S_{21}| = 0.06244$.
- $S_{11}S_{22} = 0.6982\angle -88.2^\circ$, $S_{12}S_{21} = 0.0624\angle -174.0^\circ \Rightarrow \Delta = \boxed{0.6964\angle -83.1^\circ}$, $|\Delta|^2 = 0.4850$.
- $K = \frac{1 - 0.7992 - 0.6100 + 0.4850}{0.1249} = \frac{0.0758}{0.1249} = \boxed{0.607} < 1$; $\mu = 0.863 < 1$.
- $|\Delta| < 1$ but $K < 1 \Rightarrow$ **potentially unstable**: some passive terminations make it oscillate.

Step 4: stability circles, and the proof that it is conditionally stable

Circle	Centre and radius	Nearest point to origin	Stable side
Output (Γ_L)	$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{ S_{22} ^2 - \Delta ^2} = 1.363 \angle 46.7^\circ, R_L = 0.500$	$1.363 - 0.500 = 0.863 < 1$: cuts the chart	origin outside, $ S_{11} < 1 \Rightarrow$ outside
Input (Γ_S)	$C_S = 1.132 \angle 68.5^\circ, R_S = 0.199$	$0.933 < 1$: cuts the chart	outside

- Both circles cut the unit circle, so part of the chart is unstable and the rest is stable: the device is **conditionally stable**. Pick Γ_S, Γ_L outside the circles (deck figure, §6.4).



The two circles drawn from the table: each cuts the chart, so a sliver of the chart (hatched) is unstable; the unilateral design points S_{11}^*, S_{22}^* used in Step 5 sit on the stable side. Generated from the S-parameters by `ampfig.py`, which stops if these centres and radii differ from the table.

Step 5: unilateral maximum gain, and whether its design points are safe

- $G_{TU,max} = \frac{1}{1 - 0.7992} \times 9.747 \times \frac{1}{1 - 0.6100} = 4.981 \times 9.747 \times 2.564 = \boxed{124.5 = 20.95 \text{ dB}}$.
- Design points $\Gamma_S = S_{11}^* = 0.894 \angle 60.6^\circ, \Gamma_L = S_{22}^* = 0.781 \angle 27.6^\circ$: both lie **outside** their circles, and together give $|\Gamma_{in}| = 0.912, |\Gamma_{out}| = 0.848$, both < 1 . The design is stable. **verified/added**
- But $M = 0.557$: the error bound is -3.84 to $+7.07$ dB, so 20.95 dB is only a rough estimate for this strongly bilateral device.

Step 6 (79 Bh): the bilateral comparison

- With $K < 1$ there is **no simultaneous conjugate match** and no $G_{T,max}$.
- The limit is the **maximum stable gain** $MSG = |S_{21}|/|S_{12}| = 3.122/0.020 = 156.1 = \boxed{21.9 \text{ dB}}$, approached only with stabilising loss.
- Compare: unilateral estimate 20.95 dB against an MSG ceiling of 21.9 dB.

Problem 4: A potentially unstable set, and “modify the S-parameters if necessary” 12

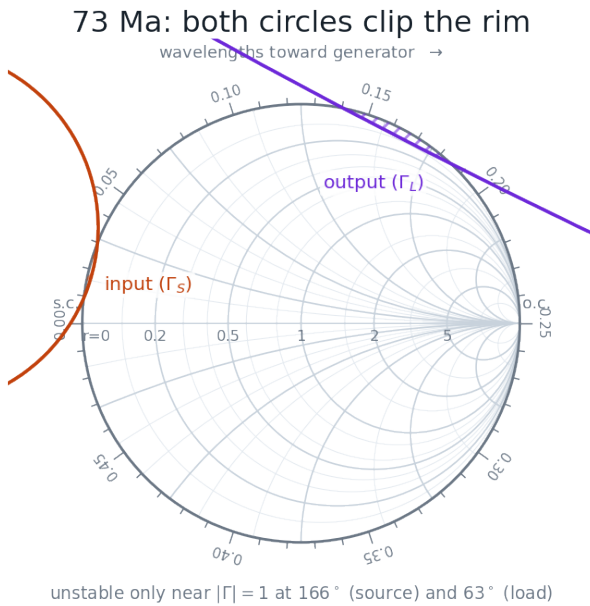
73 Ma · $|S_{21}| = 10.11$ makes this one borderline

QUESTION AS PRINTED 73 Ma [12]

Check the stability and find the maximum gain a transistor amplifier having $S_{11} = 0.64 \angle -169^\circ, S_{12} = 0.03 \angle 50^\circ, S_{21} = 10.11 \angle 91^\circ, S_{22} = 0.22 \angle -82^\circ$. Consider both bilateral and unilateral model. Modify the S-parameters if necessary.

- $|S_{12}S_{21}| = 0.3033; \Delta = 0.1985 \angle -16.9^\circ, |\Delta|^2 = 0.0394$.
- $K = \frac{1 - 0.4096 - 0.0484 + 0.0394}{0.6066} = \frac{0.5814}{0.6066} = \boxed{0.958} < 1, \mu = 0.960$: **potentially unstable**, narrowly.

- Circles: $C_S = 1.800\angle 166.0^\circ$, $R_S = 0.819$ (nearest 0.981); $C_L = 34.60\angle 62.7^\circ$, $R_L = 33.64$ (nearest 0.960). Both just clip the chart edge; stable outside. Only terminations with $|\Gamma|$ near 1 in those directions are dangerous.
- **Bilateral**: no conjugate match exists; $MSG = 10.11/0.03 = 337 = 25.3$ dB.
- **Unilateral**: $G_{TU,max} = 1.694 \times 102.2 \times 1.051 = \boxed{181.9 = 22.6 \text{ dB}}$; S_{11}^* and S_{22}^* give $|\Gamma_{in}| = 0.70$, $|\Gamma_{out}| = 0.53$: stable. $M = 0.076$, bound -0.64 to $+0.69$ dB.
- **"Modify if necessary"**: it is necessary, for the bilateral part. Worked in full after the chart.



- Both stability circles drawn on one chart, since each only **clips the rim**: the input circle near 166° , the output circle (radius 33.6, so nearly a straight line) near 63° .
- Everything inside $|\Gamma| \approx 0.96$ is stable: this is why "potentially unstable" here is narrow, and why S_{11}^* and S_{22}^* are safe.
- From `ampfig.py`, checked against the centres and radii above.

"Modify the S-parameters": stabilise with a resistor, then redo the design

- **Why**: $B_1^2 - 4|C_1|^2 = 4|S_{12}S_{21}|^2(K^2 - 1) = -0.0300 < 0$, so both roots of the conjugate-match formula have $|\Gamma_S| = 1$: no bilateral $G_{T,max}$ exists. Add a resistor so that resistor + transistor is one new two-port with $K > 1$. Its S-parameters are the "modified" ones.
- **Where**: the unstable source region hugs the rim at $\Gamma_S = 0.981\angle 166^\circ$, i.e. $Z_S = 0.49 + j6.14 \Omega$, an almost-short source. A **series** R at the gate means the transistor never sees less than R of source resistance. K belongs to the whole two-port, so once $K > 1$ both ports are safe.
- Smallest that works: $R = 0.48 \Omega$. Take $R = 5 \Omega$ for margin, $r = R/Z_0 = 0.1$.

Step A: the resistor as a two-port, cascaded in front of the transistor

- Series element z : $A_{11} = A_{22} = r/(2 + r) = 0.1/2.1 = 0.0476$; $A_{12} = A_{21} = 2/(2 + r) = 2/2.1 = 0.9524$. (Shunt y : $-y/(2 + y)$ and $2/(2 + y)$.)
- Common denominator $D = 1 - A_{22}S_{11} = 1 - 0.0476 \times 0.64\angle -169^\circ = 1 - 0.0305\angle -169^\circ = 1.0299\angle 0.32^\circ$.
- $S'_{11} = A_{11} + A_{21}^2 S_{11}/D = 0.0476 + (0.9070 \times 0.64\angle -169^\circ)/(1.0299\angle 0.32^\circ) = 0.0476 + 0.5636\angle -169.32^\circ = \boxed{0.5169\angle -168.35^\circ}$.
- $S'_{12} = A_{12}S_{12}/D = (0.9524 \times 0.03\angle 50^\circ)/(1.0299\angle 0.32^\circ) = \boxed{0.0277\angle 49.7^\circ}$.
- $S'_{21} = A_{21}S_{21}/D = (0.9524 \times 10.11\angle 91^\circ)/(1.0299\angle 0.32^\circ) = \boxed{9.349\angle 90.68^\circ}$.
- $S'_{22} = S_{22} + A_{22}S_{12}S_{21}/D = 0.22\angle -82^\circ + (0.0476 \times 0.3033\angle 141^\circ)/(1.0299\angle 0.32^\circ) = 0.22\angle -82^\circ + 0.0140\angle 140.68^\circ = \boxed{0.2099\angle -84.60^\circ}$.

These are the standard two-port cascade lines (resistor first, then transistor); `amp.py` checks them against the ABCD product.

Step B: stability of the modified device

- $|S'_{11}|^2 = 0.2672$, $|S'_{22}|^2 = 0.0441$, $|S'_{12}S'_{21}| = 0.2593$.
- $\Delta' = 0.1085\angle 107.06^\circ - 0.2593\angle 140.35^\circ = (-0.0318 + j0.1037) - (-0.1997 + j0.1655) = 0.1679 - j0.0618 = \boxed{0.1789\angle -20.2^\circ}$, $|\Delta'|^2 = 0.0320$.
- $K' = \frac{1 - 0.2672 - 0.0441 + 0.0320}{2 \times 0.2593} = \frac{0.7207}{0.5187} = \boxed{1.390} > 1$, $|\Delta'| < 1 \Rightarrow$ **unconditionally stable**.
- $|S'_{21}/S'_{12}| = 337$, unchanged (both scaled by A_{21}/D): the resistor does not touch the MSG, it raises K .

Step C: bilateral and unilateral maximum gain of the modified device

Quantity	Formula and substitution	Value
B_1, C_1	$1 + 0.2672 - 0.0441 - 0.0320$; $S'_{11} - \Delta'S'^*_{22}$	$1.1911, 0.5405\angle -165.18^\circ$
B_2, C_2	$1 + 0.0441 - 0.2672 - 0.0320$; $S'_{22} - \Delta'S'^*_{11}$	$0.7449, 0.2759\angle -69.12^\circ$
Discriminant	$4 S'_{12}S'_{21} ^2(K'^2 - 1)$	0.2504 , root 0.5004
Γ_S	$(1.1911 - 0.5004)/(2 \times 0.5405\angle -165.18^\circ)$	$0.6390\angle 165.18^\circ$
Γ_L	$(0.7449 - 0.5004)/(2 \times 0.2759\angle -69.12^\circ)$	$0.4431\angle 69.12^\circ$
$G_{T,max}$	$1.6901 \times 87.40 \times 0.9690$	$143.1 = 21.56$ dB
Check	$337.0 \times (1.3895 - 0.9648) = 337.0 \times 0.4247$	$143.1 \checkmark$
$G_{TU,max}$	$1.3646 \times 87.40 \times 1.0461$	$124.8 = 20.96$ dB
M' , bound	0.040	-0.34 to $+0.36$ dB

- **Answer:** as given, $K = 0.958$: potentially unstable, unilateral 22.6 dB, no bilateral match (MSG ceiling 25.3 dB). Modified with 5Ω in series at the gate: $K = 1.390$, bilateral $\boxed{21.56 \text{ dB}}$, unilateral $\boxed{20.96 \text{ dB}}$. The resistor costs about 1 dB and buys a design that cannot oscillate.
- Networks now present Γ_S at the resistor's outer end: $50\Omega \rightarrow$ open stub $0.1638\lambda \rightarrow$ line $0.0904\lambda \rightarrow 5\Omega \rightarrow$ gate; output stub 0.1241λ , line 0.2425λ (Problem 1's table method).
- Other placements, checked: series R at the drain needs $R > 4.02\Omega$; shunt at the drain $R < 1.76 \text{ k}\Omega$; shunt at the gate $R < 57\Omega$, which bleeds off input power. Any is accepted if you state it and recompute K .
- "Modify" can also be read as setting $S_{12} = 0$. That is only the unilateral model above; it does not make the device stable, so it cannot answer the bilateral part.

Problem 5: Gains of the input and output networks for given terminations 6+4

76 Bh

QUESTION AS PRINTED 76 Bh [6+4]

A microwave transistor at 1.0 GHz with 50Ω has $S_{11} = 0.38\angle -158^\circ$, $S_{12} = 0.11\angle 54^\circ$, $S_{21} = 3.50\angle 80^\circ$, $S_{22} = 0.40\angle -43^\circ$. The source impedance is $Z_s = 25\Omega$ and load impedance is $Z_L = 40\Omega$. Compute the gains of input and output matching networks. Also, draw a design a flowchart.

Step 1: reflection coefficients of the given terminations

$$-\Gamma_S = \frac{25 - 50}{25 + 50} = -0.3333, \quad \Gamma_L = \frac{40 - 50}{40 + 50} = -0.1111.$$

Step 2: stability first

- $\Delta = 0.2555\angle -60.6^\circ$, $K = 0.988 < 1$: potentially unstable. Check the actual terminations: $|\Gamma_{in}(\Gamma_L)| = 0.365$, $|\Gamma_{out}(\Gamma_S)| = 0.545$, both $< 1 \Rightarrow$ **stable with these terminations.** verified/added

Step 3: the three gain blocks (unilateral)

Block	Formula and substitution	Value
Input network G_S	$\frac{1 - \Gamma_S ^2}{ 1 - S_{11}\Gamma_S ^2} = \frac{1 - 0.1111}{1 - 0.1111}$	1.138 = +0.56 dB
Transistor G_0	$\frac{ 1 - 0.38\angle-158^\circ \times (-0.333) ^2}{ S_{21} ^2} = \frac{3.50^2}{3.50^2}$	12.25 = 10.88 dB
Output network G_L	$\frac{1 - \Gamma_L ^2}{ 1 - S_{22}\Gamma_L ^2} = \frac{1 - 0.0123}{1 - 0.0123}$	0.926 = -0.34 dB
Total G_{TU}	$\frac{ 1 - 0.40\angle-43^\circ \times (-0.111) ^2}{1.138 \times 12.25 \times 0.926}$	12.90 = 11.11 dB
Exact G_T (with S_{12})	full formula, Γ_{in} in place of S_{11}	12.60 = 11.01 dB

- The input termination **adds** 0.56 dB (it moves toward S_{11}^*); the output termination **costs** 0.34 dB. Maxima would be $G_{S,max} = 0.68$ dB and $G_{L,max} = 0.76$ dB. Flowchart: §6.4.

Problem 6: Designing the amplifier: matching networks and total gain 10 18

79 Ch, 69 Bh · the only questions that ask for the networks themselves

QUESTION AS PRINTED 79 Ch [10]

A GaAs power FET has the S-parameters at 5 GHz with 50-ohm line measured as $S_{11} = 0.45\angle 163^\circ$, $S_{12} = 0.04\angle 40^\circ$, $S_{21} = 2.55\angle -106^\circ$ and $S_{22} = 0.46\angle -65^\circ$. Using these parameters, design an amplifier and find the total gain power gain.

Step 1 to 4: as Problem 1 (values in Problem 2's tables, 72 Ma row)

- $K = 3.332$, $|\Delta| = 0.306$: unconditionally stable.
- $\Gamma_S = 0.4011\angle -160.6^\circ$, $\Gamma_L = 0.4131\angle 67.2^\circ$, $G_{T,max} = 1.1917 \times 6.5025 \times 1.2636 = \boxed{9.79 = 9.91 \text{ dB}}$ (total power gain).

Step 5: input matching network (open shunt stub, then a line)

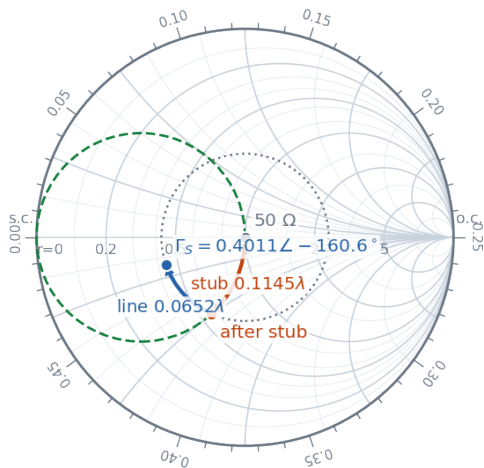
1. Start at the 50Ω source: $y = 1$, the chart centre.
2. A shunt stub moves along the $g = 1$ circle to $y = 1 + jb$; this must land on the circle $|\Gamma| = |\Gamma_S| = 0.4011$: $b = \frac{2|\Gamma_S|}{\sqrt{1 - |\Gamma_S|^2}} = 0.8757$.
3. Open stub with $b = +0.8757$: $l_{stub} = \frac{\tan^{-1} 0.8757}{2\pi} = \boxed{0.1145\lambda}$.
4. At the stub $\Gamma = \frac{1 - y}{1 + y} = 0.4011\angle -113.6^\circ$; a line of length l rotates it clockwise by $720l$ degrees to $\angle -160.6^\circ$: $l = \frac{160.6 - 113.6}{720} = \boxed{0.0652\lambda}$.

Step 6: output network, the same way

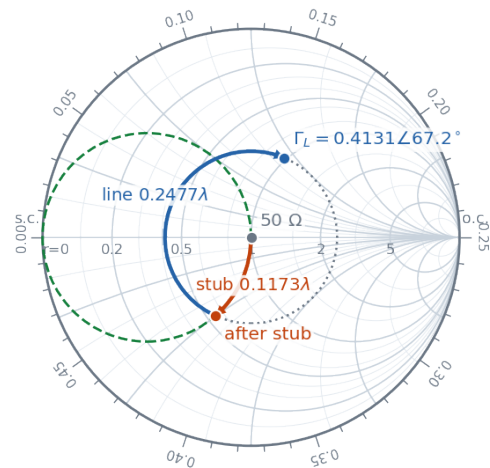
- $b = 0.9072$, open stub $\boxed{0.1173\lambda}$, line $\boxed{0.2477\lambda}$ to present $\Gamma_L = 0.4131\angle 67.2^\circ$.
- Circuit: source $\rightarrow$ stub $\rightarrow$ line $\rightarrow$ FET $\rightarrow$ line $\rightarrow$ stub $\rightarrow$ load, as in the deck's final diagram (§6.4).
- `amp.py` walks both networks forward and recovers Γ_S , Γ_L exactly.

79 Ch: input network, to Γ_S

wavelengths toward generator →

 $b = +0.8757$ (open stub), then the line rotates clockwise by 47.0° 79 Ch: output network, to Γ_L

wavelengths toward generator →

 $b = +0.9072$ (open stub), then the line rotates clockwise by 178.4°

Steps 5 and 6 on the chart. From the $50\ \Omega$ centre the open stub moves the point along the admittance $g = 1$ circle (green) onto the $|\Gamma_S|$ or $|\Gamma_L|$ circle (dotted); the line then rotates it clockwise to the required Γ . Drawn by `ampfig.py` from the same numbers, which it checks against the boxed lengths.

69 Bh [18], the same design for the 4 GHz GaAs FET

- $K = 1.195$; $\Gamma_S = 0.872 \angle 123.4^\circ$, $\Gamma_L = 0.876 \angle 61.0^\circ$, $G_{T,max} = 46.9 = 16.71\ \text{dB}$.
- Input: open stub 0.2064λ , line 0.1193λ . Output: open stub 0.2073λ , line 0.2052λ . High $|\Gamma| \rightarrow$ long stubs and a narrow bandwidth.

Problem 7: Stability from given circles, and synthesis on a sketched Smith chart 5 8 10

71 Bh, 73 Bh, 74 Bh, 75 Bh

QUESTION AS PRINTED 71 Bh [5+5]

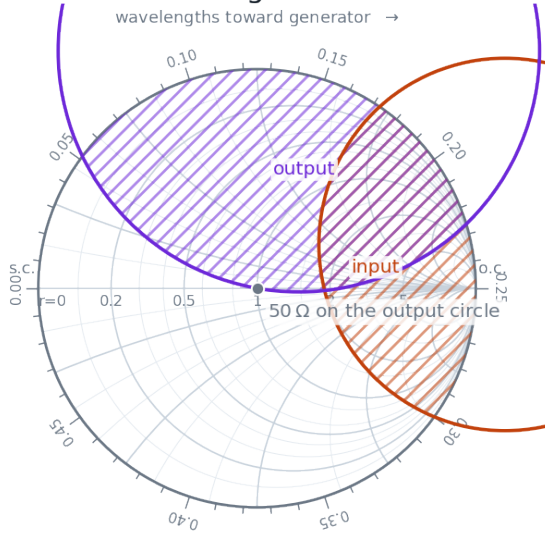
Draw a flow diagram to describe the design procedure of a microwave amplifier. Define the stability of an amplifier having $C_S = 1.15 \angle 10^\circ$, $R_S = 0.85$, $C_L = 1.10 \angle 80^\circ$, $R_L = 1.10$.

71 Bh: read the circles against the unit chart

Circle	Span of $ \Gamma $ it covers	Origin	Conclusion
Input, C_S, R_S	$1.15 - 0.85 = 0.30$ to 2.00	outside ($1.15 > 0.85$)	cuts the chart; stable outside (if $ S_{22} < 1$)
Output, C_L, R_L	$1.10 - 1.10 = 0$ to 2.20	on the circle	cuts the chart; the $50\ \Omega$ load sits exactly on the boundary

- Both circles intersect the unit Smith chart $\Rightarrow$ the amplifier is **conditionally stable**.
- The output circle passing through the centre means $\Gamma_L = 0$ gives $|\Gamma_{in}| = 1$ exactly, i.e. $|S_{11}| = 1$: a plain $50\ \Omega$ load is on the edge of oscillation. Choose Γ_L well on the stable side, away from the direction of C_L (80°). verified/added

71 Bh: the given circles



hatched = unstable side, taking $|S_{11}|, |S_{22}| < 1$ (stable outside)

- The sketch the question wants: the unit chart and the two given circles, each **cutting** it. Hatch the unstable side and say which side that is.
- The output circle passes through the centre: that is the $50\ \Omega$ load sitting on the stability boundary.
- Drawn by `ampfig.py` from the printed C and R .

73 Bh, 74 Bh, 75 Bh: synthesis from a sketched chart (assume your own numbers)

1. Read each circle's centre C (distance and angle from the chart centre) and radius R .
2. Decide the stable side by testing the centre (§6.4).
3. Mark the unstable overlap (hatch it), as in the deck's 800 MHz example: $C_S = 1.79\angle 122^\circ$, $R_S = 1.04$; $C_L = 1.30\angle 48^\circ$, $R_L = 0.45$, $K = 0.547$.
4. Choose Γ_S and Γ_L in the stable part, and for gain on the relevant constant-gain circles; confirm $|\Gamma_{in}|, |\Gamma_{out}| < 1$.
5. "Stability parameters of the input matching network" (74 Bh) = the input circle's C_S , R_S and the resulting allowed Γ_S region.

Problem 8: Unilateral design for a chosen gain, with assumed S-parameters 10

78 Ch · use the deck's 8 GHz MESFET as the assumed transistor

QUESTION AS PRINTED 78 Ch [10]

Assuming suitable S-parameters calculate maximum gain of an amplifier working in an unilateral model. Also illustrate a flow chart showing the designing steps.

Assumed: GaAs MESFET, 8 GHz, $S_{11} = 0.52\angle -145^\circ$, $S_{12} = 0.03\angle 20^\circ$, $S_{21} = 2.56\angle 170^\circ$, $S_{22} = 0.48\angle -20^\circ$

Step	Working	Value
Stability	$\Delta = 0.173\angle -162.8^\circ$; $K = \frac{1 - 0.2704 - 0.2304 + 0.0300}{0.1536}$	$K = 3.445$, stable
$G_{S,max}, G_0, G_{L,max}$	$\frac{1}{0.7296}, 2.56^2, \frac{1}{0.7696}$	1.371, 6.554, 1.299
$G_{TU,max}$	$1.371 \times 6.554 \times 1.299$	11.67 = 10.67 dB
M	$\frac{0.03 \times 2.56 \times 0.52 \times 0.48}{0.7296 \times 0.7696}$	0.0341
Bound	$1/(1.0341)^2$ to $1/(0.9659)^2$	-0.29 to +0.30 dB

- **Deck corrections:** the deck prints $|\Delta| = 0.168$, $K = 3.53$ and $M = 0.04$ (bound -0.36 to +0.37 dB); the exact values are those above. The conclusions stand. verified/added
- **For a 10 dB design (deck):** $g_S = 1.25$, $g_L = 10/(1.25 \times 6.554) = 1.221$; constant-gain circles $C_{gs} = 0.486\angle 145^\circ$, $r_{gs} = 0.222$ and $C_{gl} = 0.457\angle 20^\circ$, $r_{gl} = 0.192$; nearest points $\Gamma_S = 0.264\angle 145^\circ$, $\Gamma_L = 0.265\angle 20^\circ$ give exactly 10.0 dB; input network open stub 0.080λ + line 0.152λ , output 0.080λ + 0.326λ (deck: 0.08, 0.154, 0.082, 0.326).
- Flowchart: §6.4.

Problem 9: Negative-resistance oscillator numbers

practice from the deck (slides 64 and 66); the theory asks are 70 Ma, 71 Ma, §6.5

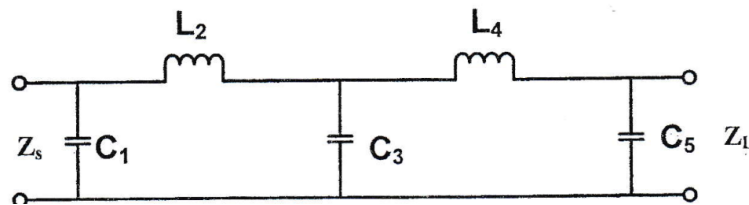
- $Z_{out} = -10\ \Omega$: $\Gamma_{out} = \frac{-10 - 50}{-10 + 50} = -1.5 = 1.5\angle 180^\circ$. Plot $1/\Gamma_{out}^* = 0.667\angle 180^\circ$, read $r = 0.2$, negate: $Z_{out} = -10\ \Omega$ ✓.
- **Gunn diode**, $\Gamma_{out} = 1.24\angle 30^\circ$ at 10 GHz: $Z_{out} = 50 \frac{1 + \Gamma}{1 - \Gamma} = -68.9 + j159.0\ \Omega$.
 - Resonate: $X_L = -159\ \Omega$, $C = \frac{1}{2\pi \times 10^{10} \times 159} = 0.100\ \text{pF}$.
 - Start-up: $R_L = |R_{out}|/3 = 23\ \Omega$ (Pozar), or at most $|R_{out}|/1.2 = 57.4\ \Omega$ by the deck's rule. The deck's $60\ \Omega$ is too large.

Problem 10: Design steps of a given fifth-order ladder, in microstrip 3+8+3

82 Bh · the figure gives topology only; specify the rest

QUESTION AS PRINTED 82 Bh [3+8+3]

Explain, why microwave filters are designed using the Insertion Loss Method? Explain the detail design steps of the following microwave filter, and implement the circuit using microstrip.



Step 1: specification (assumed, and say so)

- The ladder is shunt C_1 , series L_2 , shunt C_3 , series L_4 , shunt C_5 : a **fifth-order low-pass** filter, first element shunt.
- Assume Butterworth, $f_c = 2.5\ \text{GHz}$, $R_0 = Z_S = Z_L = 50\ \Omega$, substrate $\epsilon_r = 4.2$, $h = 1.58\ \text{mm}$.

Step 2: prototype and Step 3: scaling

Element	g_k	Scaling	Lumped value	Microstrip ($Z, \beta l, W, l$)
C_1	0.618	$g/(R_0\omega_c)$	0.787 pF	20 Ω , 14.2°, 11.27, 2.49 mm
L_2	1.618	$R_0 g/\omega_c$	5.15 nH	120 Ω , 38.6°, 0.43, 7.64 mm
C_3	2.000	$g/(R_0\omega_c)$	2.55 pF	20 Ω , 45.8°, 11.27, 8.07 mm
L_4	1.618	$R_0 g/\omega_c$	5.15 nH	120 Ω , 38.6°, 0.43, 7.64 mm
C_5	0.618	$g/(R_0\omega_c)$	0.787 pF	20 Ω , 14.2°, 11.27, 2.49 mm

$\omega_c = 2\pi \times 2.5 \times 10^9 = 1.571 \times 10^{10}\ \text{rad/s}$. Capacitor lines: $\beta l = g Z_l / R_0$; inductor lines: $\beta l = g R_0 / Z_h$. W from Hammerstad synthesis, $l = (\beta l / 360^\circ) \lambda_g$.

- **Check:** the lumped ladder gives $|S_{21}| = -3.01\ \text{dB}$ at 2.5 GHz (filt.py).
- **Step 4: implementation:** the double- π stepped-impedance layout of §6.3 (three pads, two thin lines), with $50\ \Omega$ feeds ($W = 3.13\ \text{mm}$) at both ends.
- **Why ILM** (3 marks): §6.1.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.